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Inductive Graph-based Knowledge Tracing

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***Abstract*—The rise of virtual education and increase in distance, partly owing to the spread of COVID-19 pandemic, has made it more difficult for teachers to determine each student's learning status. In this situation, knowledge tracing (KT), which tracks a student's mastery of specific knowledge concepts, is receiving increasing attention. KT utilizes a sequence of student-exercise interactive activities to predict the mastery of concepts corresponding to a target problem, recommending appropriate learning resources to students and optimizing learning sequences for adaptive learning. With the development of deep learning, various studies have been proposed, such as sequential models using recurrent neural networks, attention models influenced by transformers, and graph-based models that depict the relationships between knowledge concepts. However, they all have common limitations in that they cannot utilize the learning activities of students other than the target student and can only use a limited form of exercise information. In this study, we have applied the concept of rating prediction to the student-exercise knowledge tracing problem and solved the limitations of the existing models. Our proposed Inductive Graph-based Knowledge Tracing (IGKT) designed to integrate structural information and various unrestricted types of additional information into the model through subgraph sampling, has been found superior over the existing models across two different datasets in predicting student performances.**

***Index Terms*—knowledge tracing, graph-based knowledge tracing, IGMC, IGKT, rating prediction**

I. Introduction

In the field of education, it is a significant task to accurately evaluate the status of a student's understanding of current learning topics, and recommend tailored learning materials based on the evaluation [1]. In particular, it was reported that online teaching and virtual education, which have been adopted rapidly since the COVID-19 pandemic, caused students to be easily distracted by diverse learning environments, ultimately deepening the educational gap between the learners [2]. To overcome this gap, the demand for a tailored tutoring system is increasing, as well as a reliable method of measuring the student understanding levels based on the individual logs generated in the learning management systems (LMSs) [3].

Fig. 1. Knowledge tracing task based on student-exercise sequence.

Knowledge tracing (KT) [4] is one of the mainstream attempts to track students' learning status by analyzing the user logs from LMS. One's learning trajectory can be represented by a set of knowledge concepts (KC) and the relationships among them. KT uses a sequence of student-exercise interactions to predict the mastery of concepts corresponding to the target exercises. As shown in Figure 1, the student's understanding of a knowledge concept can be measured by referring to the testing results of whether the problems are successfully solved from the exercises related to the knowledge concept. By learning the past testing results derived from multiple knowledge concepts, predictions for unseen exercises can be made.

As KT is a prediction task in essence, the technique can get along with the learning resource recommendation [5]. In general, instructors are required to navigate through groups of students and manually select the appropriate resources for each student's level of knowledge to ensure personalized learning. Choosing exercises that are too easy or challenging hinders effective learning [6]. To avoid this problem, a model that adjusts the difficulty levels of the exercises corresponding to the student's knowledge status was proposed [6]. On the other hand, adaptive learning, which actively personalizes the learning contents or the way the contents are presented, is also opted for knowledge tracing to constantly measure the students' knowledge status [7].

KT is commonly conducted in two ways. The first is the traditional KT model, which can be represented by Bayesian

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knowledge tracing [4]. It generally uses a probabilistic model, such as the hidden Markov model [8] and Bayesian network. The second is a deep-learning-based knowledge tracing model that can be classified into three specific models: sequential models by deep knowledge tracing (DKT) [9], attentive models by self-attentive knowledge tracing (SAKT) [10], and graph-based knowledge tracing (GKT) models [11].

Despite their potential to improve personalized learning, the aforementioned approaches have focused on analyzing the interactions between single students and target exercises. This approach would be well-suited for predicting a change in an individual's learning status; however, it does not reflect other students' interactions towards similar knowledge concepts in the prediction model. In addition, the prediction cannot be generalized to cover a wide range of student learning activities.

Furthermore, these existing models do not consider additional features associated with the learning process other than correctness. Exercises in the real world consist of complex combinations of multiple knowledge concepts. However, existing methodologies either assume only one concept per problem [12], or capture the relationships between the knowledge concepts through individualized and thus fragmented perspectives only [11], overlooking the potential of utilizing the various side information of exercises and interactions together from a more comprehensive perspective.

Inspired by the user-item heterogeneous graph, which visualizes the user-item relationship as a graph in the field of recommender systems, we visualized the student-exercise pair and conducted a rating prediction for the unseen exercises by utilizing inductive graph matrix completion, as shown in Figure 2. Unlike other graph-based recommender systems that convert exercises into fixed-dimensional vectors by passing them through the embedding layer, our proposed model exploits four additional one-hot representation vectors, corresponding to the student, his or her neighbors, exercise, and exercise types, respectively. The proposed model further connects exercise nodes, where their knowledge concepts are identical. By doing so, the graph structure can be learned to represent node features. In addition, exercise features, such as knowledge concepts and exercise types, are added in the form of edges to obtain rich node representations. With this configurational arrangement, we propose a new KT model that utilizes all the learning features of other students rather than the sequence of a single student. Our experiment shows that the proposed model outperforms the existing state-of-the-art models, realizing the potential of utilizing the additional features in the graph structure.

The main contributions of our study are summarized as follows:

- We introduce an inductive student-exercise heterogeneous graph to knowledge tracing that expands from a single student learning activity to multiple student learning activities for prediction.
- We propose a method to effectively utilize the additional features of exercises in a knowledge tracing model based on a graph structure.

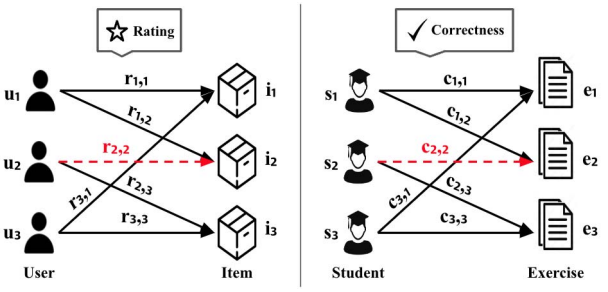

Fig. 2. Applying graph-based rating prediction to knowledge tracing

- By conducting experiments on multiple datasets, we show that a knowledge tracing task based on the proposed structure is robust.

II. Related Work

This section reviews several relevant studies that have focused on knowledge tracing models. Thereafter, graph neural network (GNN) models are introduced.

A. Knowledge tracing

As previously mentioned, there are two main KT approaches. The first is a traditional machine-learning-based approach, such as Bayesian knowledge tracing (BKT) [4]. BKT applies a hidden Markov model to tracing student skills in a binary knowledge state.

The second is a deep learning-based approach such as Deep Knowledge Tracing (DKT) [9]. DKT used RNN to update students' knowledge into a single hidden state through the student's interaction sequences. DKT was the first model to utilize DL techniques for knowledge tracing, and also showed a remarkable performance improvement. However, the model was limited in expressing the student's learning state as a single hidden state, without the distinction of specific concepts. As a result, it was difficult to discover the KCs the students understood well and the fields with which they were more comfortable [12].

The DKVMN [12], an extended version of DKT, was presented based on a memory network [13]. By using a memory matrix representing a pair of keys and values with the student's mastery for each KC, the knowledge state of the student was tracked for each concept, improving model interpretability. However, these models were also limited as they used only specific features and treated each interaction with the same weight.

To further improve the above models, the attention mechanism has been actively considered for the KT model. SAKT [10] was the first attention-mechanism-based KT model. Through the attention layers, SAKT learns the importance of each exercise in past interactions to predict whether the student will answer a new exercise correctly.

Further, SAINT [14] is a model that improved SAKT by adapting a transformer-based method including both an encoder and decoder. SAINT separates the student's interaction sequence from the exercise and response data sequence, and then provides the sequence as input to the encoder and decoder

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sequentially. SAINT+ [15] is a variant model extended by adding two features to the response embedding sequence: the time the student spent answering each question, and the time interval between two continuous learning interactions.

Also, studies have been conducted to improve the performance of knowledge tracing using various additional features. Among them, there is a trend of using graph representation learning techniques such as GNNs, which represents complex relationships such as similarity and dependency between KCs. GKT [11] is a variant of KT models that learns the relationship between KCs through a graph structure, with KCs as nodes and their dependencies as edges. However, it is different from our proposed model that can integrate various types of KCs into the model in that it still has the assumption of "one KC per question."

B. Graph neural network

Because the graph data structure is specialized in expressing the structure of user-item interaction, it has been widely used for building recommender systems in various domains. In recent years, many studies have been conducted to create user and item representations from interaction data using graph neural networks.

GCMC [16] is a transductive model designed for matrix-completion tasks. It uses a graph convolutional autoencoder to obtain the user and item representations from the rating matrix. GraphSAGE [17] is an early inductive graph framework that efficiently creates node representations from random-walk subgraphs. GraphSAGE utilizes node feature information such as text and image properties. On the other hand, subgraph-based inductive methods can be used to generalize tasks and can be applied to unseen nodes.

IGMC [18] is an inductive approach for matrix completion tasks. They extracted a local enclosing subgraph for target rating. Each subgraph consists of nodes and edges of the h-hop neighbors of each target rating. From the subgraph, the GNN learns a complex pattern of the graph and predicts its rating. In contrast to the early inductive graph frameworks, IGMC sequentially labels the nodes in the extracted subgraph based on the target rating and conducts learning through different R-GCN [19] layers according to the type of rating. Because of this structure, unlike the early inductive graph frameworks that relied on high-dimensional node features, IGMC does not rely on side information and only utilizes the graph structure and ratings.

III. PROBLEM FORMULATION

Given a target exercise, our main goal is to predict whether or not a student would give a correct answer for the exercise by learning the student's learning sequences. The length of the students' interaction sequences were varied, but we divided them into sub-sequences with fixed-length. We disallowed the overlapping between the sub-sequences. The problem formulation was defined similarly in most KT tasks [5]. Formally, the interaction sequence of a student is noted as $I = \{i_0, i_1, i_2, \ldots, i_T\}$ and it contains a sequence of exercises and its correctness. Each interaction contains exercise e_t and correctness c_t on time t. Exercise e_t contains id of the exercise and connected knowledge concepts. Correctness $c_t \in \{0, 1\}$ is equal to 1 if the student's answer is correct and 0 if it is wrong. From the sequence of the interaction history, predicting whether or not the student answers correctly is the main task of our knowledge tracing model. In terms of probability estimation, it can be expressed as shown in Equation 1.

$$P(c_t = 1 | i_0, i_1, \ldots, i_{t-1}, e_t) \tag{1}$$

IV. METHOD

In this section, we introduce the construction and learning methods of our proposed model, Inductive Graph-based Knowledge Tracing (IGKT). After introducing the overall structure of the model, each component is described in detail.

A. Overview

Figure 3 presents the three major steps of the IGKT model, each of which is detailed below:

User's sub-sequence extraction: First, the subsequence of a student's previous interactions is extracted to predict whether the student's answer is correct in the target interaction. A fixed-length interaction subsequence is extracted from the student's entire sequence immediately preceding the target interaction.

Subgraph extraction and adding exercise information: Second, the subgraph is extracted and additional information is added to it in the form of edges. The target interaction consists of the target student, the target exercise, and the subgraph constructed by extracting their 1-hop neighbors. The neighbor nodes of the target student are exercises that appear in the subsequence extracted in the first step. After subgraph construction, exercises that belong to the same part or share tags lead to a new type of edge.

Graph-based correctness prediction: Next, we derive node representations from the subgraph through relational graph convolutional network (R-GCN) layers and predict the correctness from the target student and exercise node embeddings.

B. User's sub-sequence extraction

Student subsequence extraction is the first step in the IGKT model. As previously mentioned, most knowledge tracing research defines the problem as a 'many to one' prediction task. To generalize the task, subsequences are extracted from each student's sequence, and our model is designed to learn in units of subsequences of a fixed length.

When predicting the correctness c of the interaction i_t at the time t, the prediction proceeds through a subsequence consisting of the closest n-1 interactions among the previous sequences. This subsequence-based learning method was used in previous studies [14], [15].

Existing self-attention-based models use the order information of interactions and timestamp data; however, IGKT is a model designed by focusing only on the characteristics of the graph without using such time-dependent information.

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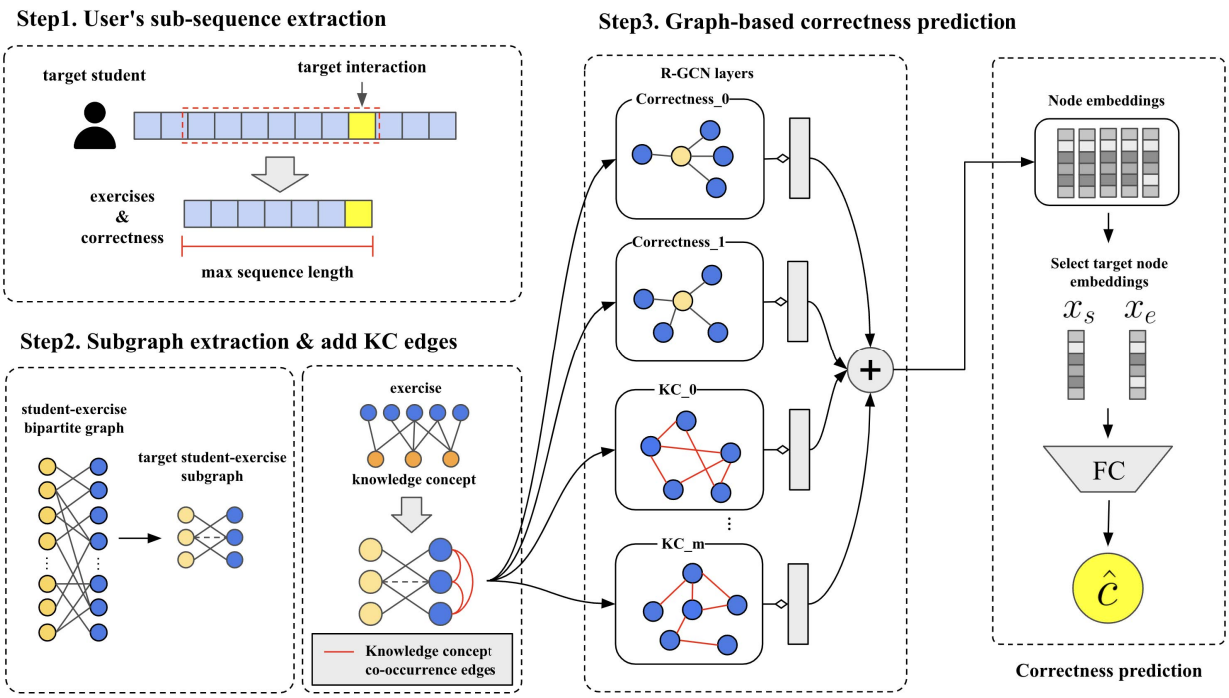

Fig. 3. The framework of inductive graph-based knowledge tracing

C. Subgraph extraction and adding exercise information

In the second step, the subgraph is extracted from the entire student-exercise graph and information about the exercise is added. When forming an entire student-exercise graph, students and exercises are converted into two types of nodes, and the interactions between students and exercises are converted into edges. The edges are classified into two types, 0 and 1, indicating whether the exercise has been correctly solved.

After forming the entire graph, a list of candidate nodes for subgraph extraction is created based on the subsequence from the previous step. The candidate node list includes the target student and n exercises included in the sequence, and $n-1$ neighboring student nodes of the target exercise are randomly selected from the entire graph and added. Thereafter, a subgraph that includes candidate nodes is extracted from the entire graph. The final extracted subgraph includes the target student, exercise, its neighboring nodes, and the interaction edges with correctness. In existing sequence-based models, only the exercise data with which the target student interacted is used; however, our method has the advantage of learning the interactions between the target exercise and other students together.

There are four types of nodes in the extracted subgraph: the target student, its neighbor exercise nodes, the target exercise, and its neighbor student nodes. Similar to IGMC, four types of one-hot vectors are assigned to each node as the initial node features. 0 and 1 are assigned to the target student and exercise, respectively. Neighboring students and neighboring exercises are assigned 2 and 3, respectively [18]. Although this graph-structure-based node feature assignment method implements inductive properties, it is difficult to utilize information about nodes included in the additional data.

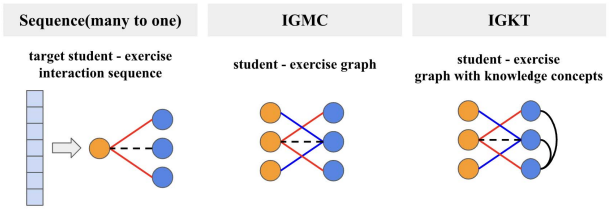

Fig. 4. Comparison of learning methods between models

In existing transformer-based KT models, each exercise ID is transformed into a fixed-dimension vector through an embedding layer. Instead, we propose a method that uses only the knowledge concepts of exercises, as illustrated in Figure 4. Because knowledge concepts such as skills and tags exist in the existing KT dataset, we connect exercises that share the same knowledge concepts.

For the EdNet dataset, each exercise belongs to one part and contains one or more tags. Additional exercise information is defined as a knowledge concept (KC), and an additional edge between exercises is created based on the co-occurrence of the KC. Table I shows the criteria for creating each new edge type. Edges can be classified into two relations: student-exercise interactions and exercise-exercise interactions.

D. Graph-based correctness prediction

In the last step, the correctness of the target interaction is predicted by passing the constructed subgraph through graph

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TABLE I
EDGE TYPES FOR EACH DATASET

Edge type	Relation	EdNet	ASSIST2017
0	student-exercise	correctness=0	correctness=0
1	student-exercise	correctness=1	correctness=1
2	exercise-exercise	same part	same problem type
3	exercise-exercise	tag co-occurrence=1	same skill
4	exercise-exercise	tag co-occurrence=2	-
5	exercise-exercise	tag co-occurrence≥3	-

neural networks. The R-GCN is applied to derive the target node representations. Equation (2) shows how the R-GCN layers aggregate neighbor node representations and update target node representations. In the equation, $x_i^{(l)}$ denotes the representations of node i from the $l-th$ layer and $\mathcal{N}_i^r$ denotes the set of neighbors of node i under relation r [19].

$$x_i^{(l+1)} = \sigma\left(W_0^{(l)} x_i^{(l)} + \sum_{r \in R} \sum_{j \in \mathcal{N}_i^r} \frac{1}{|\mathcal{N}_i^r|} W_r^{(l)} x_j^{(l)}\right) \tag{2}$$

After deriving node representations from each R-GCN layer, we concatenate the representations of node i from each layer and assign them to the final representation of node i, which is denoted by x_i' in the formula. From the final representation set, we extract the final target student representation x_s' and exercise representation x_e' using different pooling. Then, the target student and exercise representations are concatenated and the final scalar value $\hat{c}$, which represents predicted probability of correctness, is obtained through the fully connected layer. A sigmoid function is used to normalize $\hat{c}$ between 0 and 1.

$$\begin{aligned} x_i' &= concat(x_i^1, x_i^2, ..., x_i^L) \\ g &= concat(x_s', x_e') \\ \hat{c} &= Sigmoid(Wg) \end{aligned} \tag{3}$$

E. Model training

Considering that the task of our model can be considered as both binary classification and logistic regression, two loss functions are used together as shown in Equation (4) - mean squared error (MSE) and binary cross entropy loss (BCE loss). In the equation, n denotes the number of samples, c is the ground-truth correctness and $\hat{c}$ is the predicted probability of correctness. In addition, the Adam [20] optimizer is used.

$$\begin{aligned} \mathcal{L} &= \mathcal{L}_{MSE} + \mathcal{L}_{BCE} \\ \mathcal{L}_{MSE} &= \frac{1}{n}\sum_{i=0}^{n}(c_i - \hat{c}_i)^2 \\ \mathcal{L}_{BCE} &= -\frac{1}{n}\sum_{i=0}^{n}(c_i \log(\hat{c}_i) + (1-c_i)\log(1-\hat{c}_i)) \end{aligned} \tag{4}$$

V. EXPERIMENT

In this section, the experimental environment and settings are described. The proposed IGKT model was compared with other models to answer the following research question:

RQ1 Does inductive graph based knowledge tracing achieve better performance than existing models?

RQ2 Does the method of incorporating additional features using edges improve the prediction performance of the inductive graph-based approach?

RQ3 How does the performance of the model change when the length of the interaction sequence varies?

A. Dataset

The experiment was conducted using two datasets selected from different domains: EdNet [21] and ASSISTments2017Competition (ASSIST2017) [22].

EdNet comrpises English problem solving record data collected from Korean users in Santa[1], a multi-platform AI English tutoring service, and it is one of the largest public datasets in education. It has a structure with up to 6 multiple skills in one exercise problem and has a total of 7 problem types.

ASSIST2017 is a dataset of solving math problems collected from middle and high school students through the ASSISTments tutoring system[2], and it was released through the 2017 ASSISTments Data mining Competition[3]. There is one single skill per practice problem, which belongs to one of 16 problem types. Both datasets have students, exercises, skills, problem types, and timestamps. The detailed statistics of the two datasets are as follows:

TABLE II
STATISTICS OF DATASETS

Dataset	EdNet	ASSIST2017
#student	117,345	1,709
#exercise	13,517	3,162
#skill	188	102
skill type	Multiple	Single
#skill per exercise	2.292	1.00
#exercise type	7	16
#interaction per student	255.66	551.68
#interaction	30,000,000	942,816

B. Baselines

- **DKT** [9] is the model proposed by the first study that applied deep learning to knowledge tracing. It uses a recurrent neural network to track the knowledge state of students.
- **DKVMN** [12] is influenced by a memory network [13] and tracks the student's status of knowledge in the form

[1]https://www.aitutorsanta.com/kr/
[2]https://new.assistments.org/
[3]https://sites.google.com/view/assistmentsdatamining

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of an external memory matrix. Also, unlike DKT, it models students' knowledge states for individual knowledge concepts.

- **SAKT** [10] is the model proposed by the first study that applied the self-attention mechanism to knowledge tracing. It has a single encoder-based structure.
- **SSAKT** is a stack variant of SAKT and applies self-attention before inputting an exercise into a query to solve the problem of SAKT. Its performance degrades when attention blocks are stacked multiple times [14].
- **SAINT+** [15] is a model that integrates the elapsed time and lag time information into the SAINT model, which was designed to improve its performance by separating exercise and response from SAKT and learning with encoder and decoder, respectively.
- **GKT** [11] is the model proposed by the first study that applied a graph neural network to knowledge tracing. It constructs a graph by modeling the relationships between knowledge concepts, and updates the knowledge state corresponding to the related concept through the student's learning record. However, it has a limitation of modeling only one concept per exercise.
- **IGMC** [18] is a proposed model for dealing with large-scale graphs in recommender systems. It has a robust structure against changes in graph size by extracting a partial graph and shows high prediction performance without node features through only the structure information. In our implementation, only two edge types were used based on the correctness information.

C. Experimental settings

Evaluation Metric To compare the prediction performance of IGKT with other baseline methods, the *Accuracy* (ACC) and the *Area under the ROC Curve* (ROC-AUC) were used as evaluation metrics.

Train Setting All datasets were divided into training (80%) and testing (20%) based on the chronological order. In the case of the EdNet dataset, in which the exercise has up to six knowledge concepts, the upper two concepts were vectorized and integer encoding was performed for the implementation of DKT, DKVMN, and GKT. The number of preprocessed concepts is 379. Our model was implemented using Pytorch [23] and DGL [24], and the mean squared error (MSE) and binary cross entropy (BCE) loss were used together with the Adam optimizer to consider both accuracy and ROC-AUC. For a fair comparison between the models, we set the hyperparameters as identical as possible: learning rate [0.002, 0.001, 0.0005], batch size [64, 128], and number of hidden dimensions [32].

D. Experiment results

The performances of IGKT and the baselines are listed in Table III. The best performance for each dataset is indicated in bold. Our key observations made throughout the experiment are as follows.

TABLE III
ACCURACY AND ROC-AUC PERFORMANCE OF BASELINES AND IGKT

Dataset	Ednet		ASSIST2017	
Metric	ACC	ROC-AUC	ACC	ROC-AUC
DKT	0.6868	0.6771	0.6789	0.7005
DKVMN	0.6859	0.6732	0.6795	0.6952
SAKT	0.6355	0.6407	0.6575	0.6742
SSAKT	0.6466	0.6361	0.6536	0.6765
SAINT+	0.7083	0.7471	0.6807	0.7263
GKT	OOT	OOT	0.6858	0.7125
IGMC	0.6209	0.6184	0.6898	0.7572
IGKT	**0.7236**	**0.7805**	**0.7188**	**0.7919**
Improvement	2.17%	4.47%	4.21%	4.58%

The best results are highlighted in bold, and the best baseline results are indicated by underlines. Each experiment was repeated five times and *mean(±std)* is reported. "OOT" means that the experiment ran out of time.

First, IGKT exhibited the highest performance for both datasets. Among the baselines, SAINT+ for the EdNet dataset and the IGMC model for the ASSIST2017 dataset exhibited the best performance. The experiment results obtained from the two datasets showed that all models performed well on the ASSIST2017 data in terms of AUC, except for SAINT+, which was designed specifically for EdNet by the same author. This outcome is perhaps due to the fact that the data from ASSIST2017 are denser, with more than double the number of interactions per student, as shown in Table II

Second, in the case of GKT, which applied a GNN to knowledge tracing for the first time, its training time was too long to learn the EdNet dataset (out of time). GKT composes a knowledge concept dense graph that corresponds to the number of students and has a structure designed to learn node embeddings through an interaction sequence. According to the Big-O notation, it has a time complexity of $O(\#knowledge_concepts^2 * \#students * sequence_length)$ for the whole training.

Third, as an answer to **RQ1**, in the ASSIST2017 dataset, IGMC outperformed the other baseline models. The IGMC model uses subgraph sampling and structural information without additional information, suggesting that the inductive graph-based method on the student-exercise heterogeneous graph solves the knowledge tracing problem by modeling the relationship between neighboring students and exercises.

E. Ablation study

To answer **RQ2**, we performed an ablation study to verify the importance of incorporating the knowledge concepts into the knowledge tracing model. Several slim-modified models with the knowledge concepts removed were constructed for comparison with the proposed IGKT method. Each dataset has two types of knowledge concepts: skill (tag) and exercise

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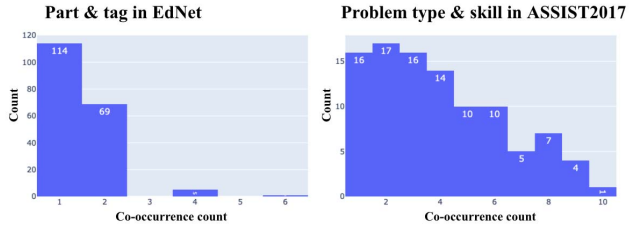

Fig. 5. Co-occurrences between different types of knowledge concepts

(part). In the main experiment (IGKT-full), both types of KC were used. However, in the slim versions, parts, tags, or both were removed. The results are presented in Table IV.

TABLE IV
ABLATION STUDY ON THE USE OF KNOWLEDGE CONCEPT IN IGKT

Dataset	Ednet		ASSIST2017	
Metric	ACC	ROC-AUC	ACC	ROC-AUC
IGMC	0.6209	0.6184	0.6898	0.7572
IGKT-part	0.7157	0.7812	0.7168	0.7861
IGKT-tag	**0.7302**	**0.7836**	**0.7298**	0.7865
IGKT-full	0.7236	0.7805	0.7188	**0.7919**

From the results on ASSIST2017 dataset, we can deduce that implementing KCs into IGMC brings significant improvements in both metrics. The large performance difference between all IGKT variants and the IGMC without KC shows that the proposed method with an edge augmentation reflects the characteristics of the problem in the graph-based approach. However, in the case of EdNet, contrary to our expectation, the performance when using only one type of KC was higher than when using both types of KC.

To investigate this issue further, we initially analyzed the two datasets in terms of tag-part overlapness. Figure 5 shows how the tag (skill) overlaps several parts (problem types) in each dataset. The x-axis represents the number of parts over which the tag is observed, and the y-axis indicates the number of tags.

In the case of EdNet, most tags appear in less than three parts. This means that the tag information is isolated according to the part. However, in ASSIST, the ratio of relatively identical tags appearing across multiple parts is high. Therefore, it can be inferred that the tag in EdNet has the characteristics of sub-information of part, whereas the skill in ASSIST contains information of a different dimension from the problem type. Through this analysis, we can infer that the relationship between the two KC types affected the performance, as both KC types in the EdNet dataset resulted in performance degradation. The observation suggests that IGKT is effective in integrating knowledge concepts independent of each other.

TABLE V
AUC PERFORMANCE RESULTS ACCORDING TO LENGTH OF SEQUENCE

Length of sequence	16	32	64	128	256
SAKT	0.6407	0.6404	0.6407	**0.6742**	0.6410
SAINT+	0.7247	0.7334	0.7471	0.7533	**0.7543**
IGMC	0.6105	0.6136	0.6184	**0.6231**	0.6171
IGKT	**0.7828**	**0.7828**	0.7805	0.7808	0.7620

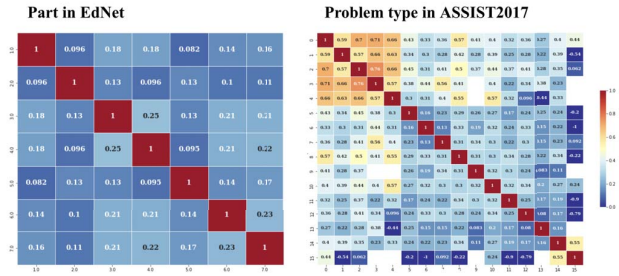

Fig. 6. Heatmaps showing correlations of percent correct with other knowledge concepts in the datasets

F. Analysis

1) Effect of sequence length

To answer **RQ3**, we traces the performance changes with various maximum sequence lengths for each model.

As shown in Table V, SAKT and SAINT+ using the attention mechanism showed performance improvement as the sequence increased. However, the performance of SAKT decreased when the length was increased to 256, probably because SAKT does not use additional time-related features and cannot filter outdated data.

IGKT achieved the highest performance at lengths of 16 and 32, and IGMC achieved the highest performance at a length of 128. We interpret this finding from two perspectives: First, IGKT exhibited a decrease in performance as the sequence increased. This outcome is attributed to the fact that graph-based models have a structure that learns with the same weights for previous interactions regardless of the chronological order. Second, IGMC achieved the highest performance in a longer sequence because it requires more correctness sequence information. It only uses overall correctness without considering the relationships between exercises. Nevertheless, it is encouraging that our proposed model outperforms existing transformer-based models in terms of the time sequence, even in short sequences, suggesting that higher performance can be achieved if a component that considers the time order is added to the graph-based method.

2) Associations between knowledge concepts

On the other hand, in EdNet, the correlations between parts are relatively low. It means that most students have different probabilities of correctness for each part, and it can be interpreted that the knowledge concepts represented by each part are isolated. Thus, we infer that the performance differences between IGMC and IGKT were larger in EdNet in the main experiment. If part information representing the isolated KC is not used, the level difference for each KC of the student cannot be captured. In IGKT, we achieved higher

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performance by solving this problem by adding edges that represent KC relevance. Figure 6 shows the correlation of the correctness between the parts of each dataset as a heatmap. The closer to blue, the lower the correlation. We observed that the ASSIST dataset had higher inter-part correlations than EdNet. In the ASSIST dataset, the correctness of other parts can be predicted from the correctness of one part.

VI. Conclusion

In this study, we have proposed IGKT, which utilizes student-exercise interaction and knowledge concept information in an inductive graph-based model. We have expanded the existing method of using only the interaction sequence of the target student to utilizing the interaction information of other students. The edge representing the KC was added to learn the association between exercises. Through the R-GCN layer, the target student and exercise representation vectors were obtained from the subgraph, and the probability of correctness was predicted.

In our experiment, we have demonstrated that the proposed model is superior to the state-of-the-art knowledge tracing models, such as SAINT+. We have confirmed that the proposed model is superior to the existing graph-based KT models in terms of performance and time complexity. In addition, our proposed method of adding KC as an edge has been found helpful for performance improvement, and the conditions for improving performance using multiple types of KC have been explicated.

In our future work, we plan to add sequence-aware components to our model. To utilize only the necessary information in a longer sequence, we will need to study a GNN-based KT model that can utilize the order and timing of interactions. Through this study, it is expected that it will be possible to more accurately capture the knowledge level of students that changes over time and to expand it to various applications that can increase learning efficiency. The current study represents a modest yet meaningful stepping stone toward this direction.

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